

3.1 Definition through linear combinations

Nearly every exact distributional statement in the first half of this book has the same form. The response is $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with normal errors, and the object of interest is a linear function of $\mathbf{Y}$ (a coefficient estimate, a fitted value, a residual) or a quadratic function of $\mathbf{Y}$ (a sum of squares). A usable notion of normality for random vectors must therefore do two things. It must survive every linear map, including maps that collapse dimensions. And it must say when two linear functions of the same vector are independent.

The residual vector shows why the first requirement has to include singular cases. In a model with p linearly independent columns its covariance matrix has rank $n - p$ (Chapter 6), so the residuals have no density on $\mathbb{R}^n$. Defining multivariate normality through a density would exclude it. This chapter uses a definition that does not mention densities at all: a random vector is normal when every linear combination of its entries is normal. The density, the moment generating function and the representation $\boldsymbol{\mu} + \mathbf{A}\mathbf{Z}$ all follow as theorems.

3.1.1 The univariate normal, including the degenerate case

For $\mu \in \mathbb{R}$ and $\sigma^2 > 0$, a random variable X has the $N(\mu, \sigma^2)$ distribution when it has density $(2\pi\sigma^2)^{-1/2} \exp\{-(x - \mu)^2/(2\sigma^2)\}$. We also allow $\sigma^2 = 0$, and take $N(\mu, 0)$ to be the point mass at μ . Treating constants as normal variables with zero variance costs nothing and removes a great many special cases later.

Lemma 3.1.1

Let $X \sim N(\mu, \sigma^2)$ with $\sigma^2 \geq 0$.

- (a) $E e^{sX} = \exp(\mu s + \frac{1}{2}\sigma^2 s^2)$ for every real s .
- (b) A random variable whose moment generating function is $\exp(\mu s + \frac{1}{2}\sigma^2 s^2)$ on an interval around 0 has the $N(\mu, \sigma^2)$ distribution.
- (c) $aX + b \sim N(a\mu + b, a^2\sigma^2)$, and if $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ are independent, then $X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Proof.

- (a) For $\sigma^2 = 0$ this is $e^{\mu s}$. Otherwise, completing the square, $sx - (x - \mu)^2/(2\sigma^2) = \mu s + \sigma^2 s^2/2 - (x - \mu - \sigma^2 s)^2/(2\sigma^2)$, and the remaining Gaussian integral equals $(2\pi\sigma^2)^{1/2}$.
- (b) is the uniqueness theorem for moment generating functions (Theorem 2.6.1(a)), in its scalar form.
- (c) The moment generating function of $aX + b$ is $e^{bs} E e^{(as)X} = \exp\{(a\mu + b)s + \frac{1}{2}a^2\sigma^2 s^2\}$. For the sum, independence gives $E e^{s(X_1 + X_2)} = E e^{sX_1} E e^{sX_2}$, which is the normal moment generating function with the added parameters. Apply (b). ■

3.1.2 The definition

Definition 3.1.1 — Multivariate normal distribution

A random vector $\mathbf{Y} = (Y_1, \dots, Y_n)^\top$ has a **multivariate normal distribution** if $\mathbf{a}^\top \mathbf{Y}$ has a univariate normal distribution (possibly degenerate) for every $\mathbf{a} \in \mathbb{R}^n$. Such a $\mathbf{Y}$ is also said to be **jointly normal**, and its entries are said to be jointly normal random variables.

Taking $\mathbf{a}$ to be a coordinate vector shows that each Y_i is normal, so $E Y_i^2 < \infty$. By the Cauchy–Schwarz inequality every product $Y_i Y_j$ is integrable, so the mean vector $\boldsymbol{\mu} = E \mathbf{Y}$ and the covariance matrix $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y})$ of Definition 2.1.2 exist. By Theorem 2.2.1, $E(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\mu}$ and $\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$, and a normal variable is determined by its mean and variance. Hence

$$\mathbf{a}^\top \mathbf{Y} \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}) \quad \text{for every } \mathbf{a} \in \mathbb{R}^n. \quad (3.1.1)$$

We write $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. The notation claims that the joint distribution of $\mathbf{Y}$ depends only on $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. That is true, but it is not obvious: the definition fixes the law of each one-dimensional projection, and it is not yet clear that these laws pin down the joint law. The claim is proved in Theorem 3.2.1. Until then, $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ should be read as “normal, with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$ ”.

3.1.3 Existence

Theorem 3.1.2 — Existence and the representation $\boldsymbol{\mu} + \mathbf{A}\mathbf{Z}$

Let $Z_1, \dots, Z_m$ be independent $N(0, 1)$ variables and $\mathbf{Z} = (Z_1, \dots, Z_m)^\top$.

- (a) $\mathbf{Z} \sim N_m(\mathbf{0}, \mathbf{I}_m)$.
- (b) For any $n \times m$ matrix $\mathbf{A}$ and $\boldsymbol{\mu} \in \mathbb{R}^n$, $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{A}\mathbf{Z}$ is multivariate normal with mean $\boldsymbol{\mu}$ and covariance $\mathbf{A}\mathbf{A}^\top$.
- (c) For every $\boldsymbol{\mu} \in \mathbb{R}^n$ and every nonnegative definite $n \times n$ matrix $\boldsymbol{\Sigma}$ there is a random vector with the $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ distribution, for example $\boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2} \mathbf{Z}$ with $m = n$.

No other matrices occur: the covariance matrix of a normal vector is nonnegative definite.

Proof.

- (a) $\mathbf{a}^\top \mathbf{Z} = \sum_i a_i Z_i$ is a sum of independent normal variables $a_i Z_i \sim N(0, a_i^2)$, which is normal by Lemma 3.1.1(c).
- (b) $\mathbf{a}^\top \mathbf{Y} = \mathbf{a}^\top \boldsymbol{\mu} + (\mathbf{A}^\top \mathbf{a})^\top \mathbf{Z}$ is a constant plus a linear combination of $\mathbf{Z}$, hence normal by (a). Its moments come from Theorem 2.2.1: $E \mathbf{Y} = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{Y}) = \mathbf{A} \mathbf{I} \mathbf{A}^\top = \mathbf{A} \mathbf{A}^\top$.
- (c) By Theorem 1.7.6 a nonnegative definite $\boldsymbol{\Sigma}$ has a symmetric nonnegative definite square root with $\boldsymbol{\Sigma}^{1/2} \boldsymbol{\Sigma}^{1/2} = \boldsymbol{\Sigma}$. Take $\mathbf{A} = \boldsymbol{\Sigma}^{1/2}$ in (b). The last sentence is Theorem 2.2.3.

Any factorization $\Sigma = \mathbf{A}\mathbf{A}^\top$ serves in (b). The symmetric square root is convenient in proofs. The Cholesky factor, a lower triangular $\mathbf{L}$ with $\mathbf{L}\mathbf{L}^\top = \Sigma$, is cheaper to compute and is what software uses. A factor with fewer columns than rows is natural when Σ is singular. Once Theorem 3.2.1 is proved, all of these give the same distribution. The first listing checks this by simulation. It draws 400,000 vectors from a three-dimensional normal distribution with each of two factors. Both sample covariance matrices agree with Σ to within 0.016 in every entry, and the quantiles of a linear combination match the exact normal quantiles.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(301)
mu = np.array([1.0, -2.0, 0.5])
Sigma = np.array([[4.0, 1.2, -0.8],
                  [1.2, 2.0, 0.6],
                  [-0.8, 0.6, 1.5]])
N = 400_000

def draw_normal(mu, A, size, rng):
    """Rows are independent copies of mu + A Z with Z ~ N(0, I)."""
    Z = rng.standard_normal((size, A.shape[1]))
    return mu + Z @ A.T

L = np.linalg.cholesky(Sigma)                # lower triangular, L L^T = Sigma
lam, U = np.linalg.eigh(Sigma)
root = U @ np.diag(np.sqrt(lam)) @ U.T       # symmetric square root

Y_chol = draw_normal(mu, L, N, rng)
Y_root = draw_normal(mu, root, N, rng)
for Y in (Y_chol, Y_root):
    print(np.round(np.cov(Y, rowvar=False) - Sigma, 3))
```

[Run this code online](#)

3.1.4 Singular normal distributions

When Σ is singular, Definition 3.1.1 still applies, and the distribution is called **singular** or degenerate. If $\Sigma\mathbf{v} = \mathbf{0}$, then $\text{Var}(\mathbf{v}^\top \mathbf{Y}) = \mathbf{v}^\top \Sigma \mathbf{v} = 0$, so $\mathbf{v}^\top \mathbf{Y} = \mathbf{v}^\top \boldsymbol{\mu}$ with probability one. Every vector in $\mathcal{N}(\Sigma)$ gives such a linear constraint, and together they say that $\mathbf{Y}$ lies in the affine subspace $\boldsymbol{\mu} + \mathcal{C}(\Sigma)$ with probability one (Theorem 2.2.3). A subspace of dimension $\text{rank}(\Sigma) < n$ has Lebesgue measure zero, so no density on $\mathbb{R}^n$ is possible.

Example. (Three overlapping sums)

Let $\mathbf{Z} \sim \mathcal{N}_3(\mathbf{0}, \mathbf{I})$ and $\mathbf{U} = (Z_1 + Z_2, Z_2 - Z_3, Z_1 + Z_3)^\top$. Then $\mathbf{U} = \mathbf{A}\mathbf{Z}$ with rows $(1, 1, 0)$, $(0, 1, -1)$ and $(1, 0, 1)$, so $\mathbf{U}$ is normal with mean zero and covariance

$$\mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}.$$

The vector $(1, -1, -1)^\top$ lies in $\mathcal{N}(\mathbf{A}\mathbf{A}^\top)$, and the rank is 2. Correspondingly $U_1 - U_2 - U_3 = 0$ for every outcome, not merely with probability one. Each U_i is $N(0, 2)$, each pair (U_i, U_j) has a nonsingular covariance matrix (determinant 3) and a bivariate density, and yet the triple lives on a plane.

In regression the singular case is the rule. With $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$, the fitted values and the residuals are normal vectors of ranks $\text{rank}(\mathbf{X})$ and $n - \text{rank}(\mathbf{X})$ (Example (Least squares under normal errors)). The last part of the listing draws a normal vector in $\mathbb{R}^3$ from a 3×2 factor. Every draw lies on the plane $\boldsymbol{\mu} + \mathcal{C}(\mathbf{A})$ to within 2.7×10^{-15} , which is rounding error.

3.1.5 Why linear combinations suffice

[Definition 3.1.1](#) describes a joint distribution entirely through its one-dimensional projections. That this loses no information is a general fact about random vectors, the Cramér–Wold device proved with characteristic functions in Proposition 2.6.2: if $\mathbf{a}^\top \mathbf{Y}$ and $\mathbf{a}^\top \mathbf{W}$ have the same distribution for every $\mathbf{a} \in \mathbb{R}^n$, then $\mathbf{Y}$ and $\mathbf{W}$ have the same distribution. We do not need it in this generality. For normal vectors the moment generating function is finite everywhere, and the next section runs the same argument with it. The device explains why the definition is natural. It also warns that all directions $\mathbf{a}$ matter. Normality of the n coordinate variables, which are only n directions, is not enough ([Section 3.3](#) and Exercise (B3)).

3.1.6 Exercises

3.1.6.1 A. Check your understanding

Exercise A1.

Let $Y_1, \dots, Y_n$ be independent with $Y_i \sim N(\mu_i, \sigma_i^2)$. Show directly from [Definition 3.1.1](#) that $\mathbf{Y} = (Y_1, \dots, Y_n)^\top$ is multivariate normal, and find its mean and covariance. Show also that a constant vector $\mathbf{c}$ is $N_n(\mathbf{c}, \mathbf{0})$.

Exercise A2.

For the bivariate normal of Example (The bivariate normal), find the Cholesky factor $\mathbf{L}$ of $\boldsymbol{\Sigma}$ and write Y_1 and Y_2 in terms of two independent standard normals. Do the same with the symmetric square root when $\sigma_1 = \sigma_2 = 1$. Why do the two representations define the same distribution?

3.1.6.2 B. Practice

Exercise B1.

Let $Z \sim N(0, 1)$ and $\mathbf{Y} = (Z, 2Z - 1, 3 - Z)^\top$. Show that $\mathbf{Y}$ is normal, find $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$, and find the rank of $\boldsymbol{\Sigma}$. Describe the set in which $\mathbf{Y}$ lies, and write it as the solution set of two linear equations.

Solution. $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{A}Z$ with $\boldsymbol{\mu} = (0, -1, 3)^\top$ and $\mathbf{A} = (1, 2, -1)^\top$, so $\mathbf{Y}$ is normal by [Theorem 3.1.2](#) with

$$\boldsymbol{\Sigma} = \mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ -1 & -2 & 1 \end{pmatrix}, \quad \text{rank}(\boldsymbol{\Sigma}) = 1.$$

$\mathbf{Y}$ lies on the line $\{\boldsymbol{\mu} + s\mathbf{A} : s \in \mathbb{R}\}$. The null space of $\boldsymbol{\Sigma}$ is $\mathbf{A}^\perp$, spanned by $(2, -1, 0)^\top$ and $(1, 0, 1)^\top$. The corresponding constraints $\mathbf{v}^\top \mathbf{Y} = \mathbf{v}^\top \boldsymbol{\mu}$ are $2Y_1 - Y_2 = 1$ and $Y_1 + Y_3 = 3$, and both hold identically.